

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 23rd February, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 3	CLOs to be covered: CLO1
Lab Title: While Loops	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Mark s	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 3: While Loops

Lab Goals/Objectives:

To understand and apply the while loop construct in Python to repeat a block of code as long as a given condition remains true.

Equipment Required:

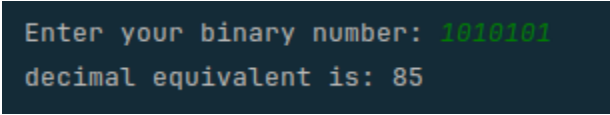
Python 3, VS Code / PyCharm

Theory:

The while loop repeatedly executes a block of statements as long as a given condition evaluates to True. It is typically used when the number of iterations is not known in advance.

Lab Work:**Task 1: Binary to Decimal Equivalent**

```
bin = (input("Enter your binary number: "))
decimal=0
power=0
i=len(bin)-1
while i>=0:
    decimal+=int(bin[i])*(2**power)
    power+=1
    i-=1
print(f"decimal equivalent is:",decimal)
```

Result:

```
Enter your binary number: 1010101
decimal equivalent is: 85
```

Task 2: Grades (While)

```
students = int(input("Enter number of students: "))
i=1
while i<=students:
    name = input("Enter name: ")
    roll_no = int(input("Enter roll number: "))
    obt_marks = int(input("Enter your marks: "))
    total_marks = int(input("Enter your total marks: "))
    print("Name :",name)
    print("RollNo :",roll_no)
    print("Obtained :",obt_marks)
    print("Total :",total_marks)
    if obt_marks < total_marks:
        percentage = (obt_marks / total_marks)* 100
        print (percentage)
        if percentage >= 90:
```

```

        print("A")
    elif percentage >= 85:
        print("A-")
    elif percentage >= 80:
        print("B+")
    elif percentage >= 75:
        print("B")
    elif percentage >= 70:
        print("B-")
    elif percentage >= 65:
        print("C+")
    elif percentage >= 60:
        print("C")
    elif percentage >= 55:
        print("C-")
    elif percentage >= 50:
        print("D")
    elif percentage < 50:
        print("F")
    elif obt_marks > total_marks:
        print ("Not Valid")
    elif total_marks == 0:
        print ("Not Defined")
    elif obt_marks == total_marks:
        print ("100%")
    i=i+1

```

Result:

```

Total Subject are 2
Enter Grade Points: 4
Enter credit hours: 3
Enter Grade Points: 3
Enter credit hours: 2
Your credit hours are 5
Your grade points are 18
Your GPA for this semester is 3.6

```

Conclusions:

The while loop was practiced through number-conversion and condition-controlled iteration problems.